



Soil layering and preferential flow impacts on seasonal changes of GPR signals in two contrasting soils

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ABSTRACT

Experimental evidence has shown that soil layering can significantly alter water movement through soil profiles, especially in sloping landscapes. Detailed knowledge of soil layering and its position in the landscape is therefore necessary for determining dynamic subsurface flow. In this study, we used Ground Penetrating Radar (GPR) in combination with high resolution real-time soil water monitoring to investigate seasonal changes in GPR signals to identify soil horizon interfaces and their impacts on water flow in two contrasting soils. The results showed that 1) in the deep Rushtown soil located in a concave hillslope, GPR reflections in the Bw–BC and BC–C horizon interfaces became clearer as soil became wetter. Such increased contrast along interfaces was due to the presence of water associated with water-restricting BC horizon and lateral flow from upslope area; and 2) in the shallow Weikert soil located in a planar hillslope, GPR reflections in the soil–bedrock interface and the weathered–unweathered rock interface become intermittent as soil became wetter. This scattering of GPR reflection was due to the non-uniform distribution of water in fractured shale bedrock and possible tree root effects. While wet condition permitted the enhanced detection of water distribution in both soil profiles, optimal time for detecting soil layering was different for these two soils: wet condition was more suited for detecting water-restricting dense layer in the deep Rushtown soil, while dry season was optimal for identifying the soil–bedrock interface in the shallow Weikert soil. Overall, this study demonstrates the value of repeated GPR surveys in different seasons to capture subsurface hydrologic processes and the impacts of soil layering and preferential water distribution on seasonal changes of GPR signals. It also provides a useful reference for selecting optimal time for GPR survey in the field.

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1. Introduction

Subsurface water movement at the pedon and hillslope scales is an important process that contributes substantially to runoff (Bonell, 1998; Hewlett and Hibbert, 1965; Lin, 2010; Weyman, 1973; Whipkey, 1965). Among many factors that influence subsurface water flow, soil layering is a significant one because changes in soil texture, density, or structure across the boundary of two adjacent layers could considerably alter water flow direction and soil moisture distribution. For instance, Brown et al. (1999) found rapid delivery of water through O-horizon above the mineral soil to stream during summer rainstorms through tracer experiment. By using crest piezometers along slope, Todd et al. (2000) identified the formation of perched water table at the clay rich B horizon during rainfall events, and these transient perched water tables led to saturated downslope flow through more permeable upper soil layer. Lin and Zhou (2008) identified preferential flow from overlying shallow soil to fractured bedrock

underneath, even during dry season when storms occurred. Many researchers (e.g., Freer et al., 2002; Tromp-van Meerveld and McDonnell, 2006; Weiler and McDonnell, 2004) have also observed rapid lateral flow at the soil–bedrock interface. As a consequence, detailed knowledge of soil layering and its spatial distribution is helpful to determining subsurface flow. However, traditional methods used to study subsurface flow have been largely destructive and/or based on point measurements, which are both labor intensive and lack of information on spatial distribution pattern.

Ground penetrating radar (GPR) is a high-resolution geophysical technique that utilizes the transmission and reflection of high frequency (generally 10–1200 MHz) electromagnetic waves to detect subsurface features nondestructively. Application of GPR relies on contrasting electrical impedance of the media being investigated, which is primarily determined by dielectric and conductive properties of the material, its magnetic permeability, and its physical properties (Daniels, 2004). For most natural soils, differences in water content are the main factor impacting GPR reflections at different times because the dielectric constant of water is about 80 while that of dry soil is between 3 and 5 (Topp et al., 1980). Such contrast between water and dry soil makes it possible to detect seasonal change in GPR signals induced by changing water

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content, especially at the interface of different soil layers that tends to cause more contrasting water distribution.

Ground penetrating radar has been used widely to study soils, including soil horizon detection and soil water content determination. For example, Gish et al. (2002) used GPR to map the distribution of restricting soil layer and combined this information with soil moisture data to infer subsurface flow pathway. Lunta et al. (2005) used GPR to detect reflection of subsurface layer with known depth and use this information to estimate soil water content and showed a comparable accuracy between GPR estimated soil water content and those determined from conventional methods such as neutron probe and TDR. Grote et al. (2005) used GPR to detect the reflection of a pavement with known depth and demonstrated that GPR can be used to monitor soil water content in sub-asphalt layers. Gerber et al. (2007) applied GPR to detect Pleistocene periglacial slope deposit (PPSD) in a test site with different parent materials and their results indicated that the condition at the test site may be applied to the field identification of PPSD. Truss et al. (2007) used time-lapsed GPR surveys to track vertical water movement in a karstic dissolution hole and compared GPR signals in dry and wet conditions to identify water distribution in oolite rock. Poisson et al. (2009) used GPR and other geophysical tools to image internal structure and water infiltration in a waste rock pile and showed that preferential flow occurred toward oxidized horizon in the east of their test site that had a fine particle size and high water retention capacity.

Combining repeated seasonal GPR surveys with real-time soil moisture monitoring to detect subsurface hydrologic processes and related influences from soil layering and preferential flow on GPR signal changes has not yet been reported. Time-lapse GPR offers insight into flow dynamics and pathways, while the combined use of GPR with soil moisture sensors allows the confirmation and extension of point-based measurements to larger areas (Freeland and Odhiambo, 2006; Saintenoy et al., 2008; Trinks et al., 2001). However, short duration time-lapsed GPR investigations (such as that done with artificial infiltration experiments) could not consider effects from seasonal hydrologic processes. Seasonal GPR signal changes under natural conditions would permit a broader understanding of the effects of rainfall, evapotranspiration, and soil moisture dynamics on subsurface hydrologic processes. This is because seasonal soil moisture dynamics would affect GPR reflections, which could be linked to soil layering and preferential water distribution. By combining seasonal time-lapse GPR surveys with high resolution real-time soil moisture monitoring, subsurface flow dynamics and flow paths can be determined. Therefore, the objectives of this study were to 1) investigate the effect of seasonal soil water dynamics on GPR signal change, especially at the identified soil horizon interfaces, and 2) reveal subsurface flow in two contrasting soils located in two different hillslopes in a forested catchment. By so doing, we hope to prove the value of combining seasonal GPR surveys with real-time soil moisture monitoring to nondestructively reveal subsurface flow dynamics and patterns and to identify optimal time for using GPR to detect soil and/or rock layers in different site conditions.

2. Materials and methods

2.1. Study site and soils

This study was conducted in the Shale Hills Critical Zone Observatory located in central Pennsylvania, which is a 7.9-ha headwater forested catchment (Fig. 1). The soils in this catchment were formed from shale colluvium or residuum and have a dominant texture of silt loam in the surface layers with silty clay loam in deeper soil layers (Table 1).

Small grids were established on two contrasting soils on two hillslopes to compare temporal variations in GPR reflection signals (Fig. 1). The first grid was located in the middle of a concave hillslope (swale), with the soil identified as the Rushtown series (loamy-skeletal over fragmental, mixed, mesic Typic Dystrochrepts, according to the U.S. Soil Taxonomy). These are deep soils with over 2 m depth to bedrock

and local slope of about 15%. The second grid was located in the middle of a planner hillslope, with the soil identified as the Weikert series (loamy-skeletal, mixed, active, mesic Lithic Dystrudepts). These are shallow soils with depth to bedrock less than 0.5 m and local slope of about 30%. Table 1 lists the basic features of these two soils. Additional details about the soils studied can be found in Lin (2006).

2.2. Experiment setup

A 4.5 by 3.7 m survey grid, perpendicular to the downslope direction, was established in the Rushtown soil (Fig. 1). For accurate repeatability of survey locations, GPR survey lines were scanned along a fixed rope that was anchored in the ground by PVC tube installed permanently at the starting and ending points of the marked survey lines. The interval of survey lines was 0.31 m. Because of limited open area restricted by trees, a smaller grid of 2 by 0.7 m, also perpendicular to the downslope direction, was established in the Weikert soil (Fig. 1). Fixed ropes were used to guide GPR scans to ensure the repeatability of locations, and the interval of the survey lines was 0.1 m in the Weikert soil.

To understand the effect of seasonal water dynamics on GPR signal, the two grids were located close to two permanently installed automatic soil water moisture monitoring stations (Fig. 1). When establishing these monitoring stations in 2006, a soil pit was excavated in each soil and an array of sensors, as shown in Fig. 1, was installed in different soil horizons and selected interfaces. These sensors include soil moisture content EC-5 (Decagon Devices, Inc., Pullman, WA) and TE525-WS tipping-bucket rain gauges (Texas Electronics Inc., Dallas, TX), which were all connected to a CR-10× datalogger (Campbell Scientific Inc., Logan, UT). The monitoring data were collected at a 10-minute time interval.

2.3. GPR instrument and survey

The TerraSIR Subsurface Interface Radar (SIR) System-3000 (Geophysical Survey Systems, Inc., Salem, New Hampshire) with a 400-MHz antenna was used in this study. The 400-MHz was chosen after in situ testing with a series of antenna frequencies (ranging from 200 to 900 MHz). The 400-MHz frequency provided the optimal balance of image quality, detection depth, and the convenience of operation in sloping landscape. Four GPR surveys in each of the two soils were conducted on October 20th and November 10th of 2008 and April 24th and May 27th of 2009.

All GPR data collected were processed using the GPR-SLICE version 7.0 software (Geophysical Archaeometry Laboratory, Woodland Hills, California). All radargrams were processed in exact same way, including the following basic steps: (1) drift removal of direct current voltage to remove signal drift from zero value; (2) zero time adjustment to find true ground surface reflection; and (3) same gain factor applied for consistent comparison.

2.4. Subsurface layer identification

The strength of GPR reflection is a function of the contrast in dielectric constant across a reflecting boundary, and can be quantified by reflection coefficient r as:

$$r = \frac{\sqrt{\epsilon_U} - \sqrt{\epsilon_L}}{\sqrt{\epsilon_U} + \sqrt{\epsilon_L}}, \quad (1)$$

where ϵ_U and ϵ_L are the dielectric constants for the upper and lower soil layers, respectively. Reflection coefficient, indicating the contrast between two adjacent layers, has a range between 1 and -1 . Positive sign indicates a higher dielectric constant in the upper soil horizon and negative sign indicates a lower dielectric constant in the upper horizon. A higher absolute value of reflection coefficient indicates a higher

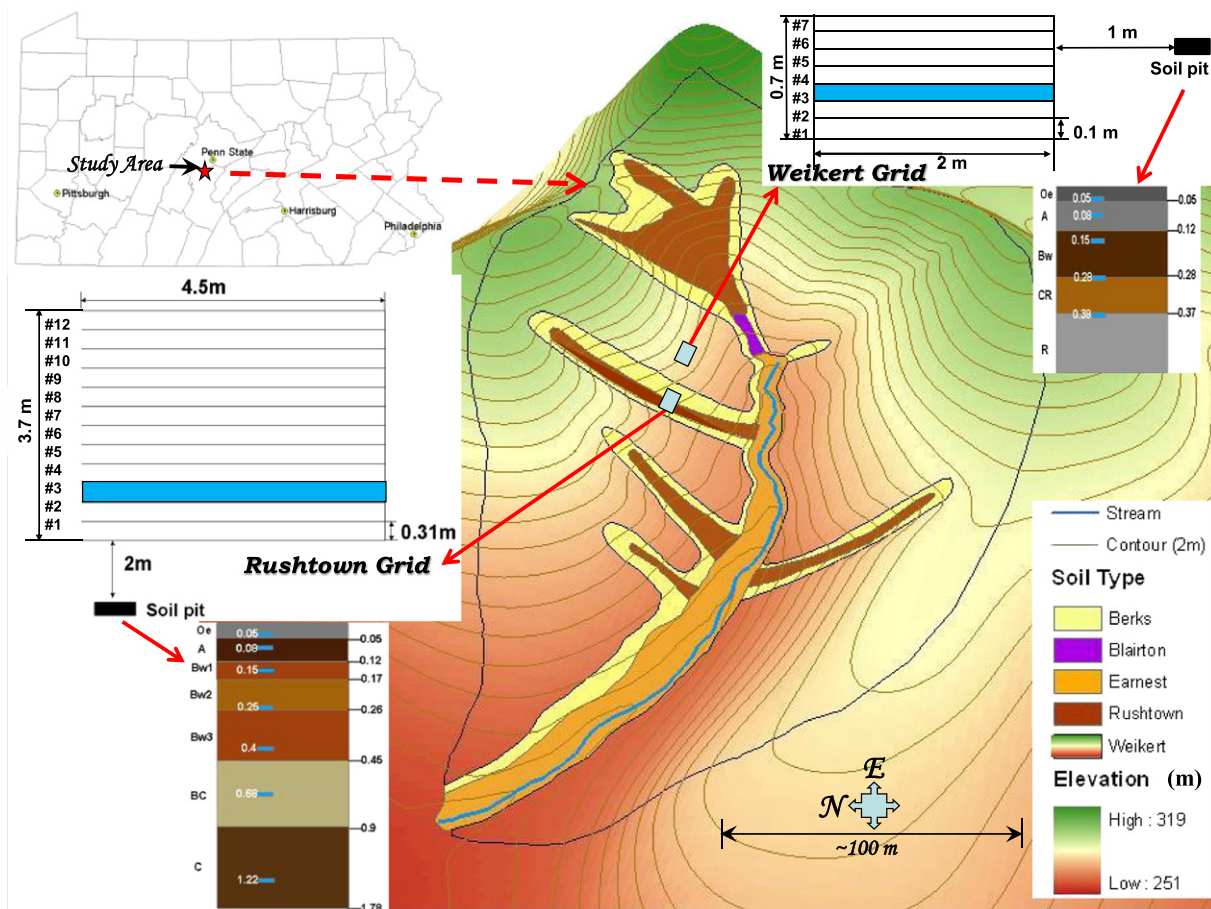


Fig. 1. Locations of GPR surveys in two contrasting soils and hillslopes in the Shale Hills Catchment located in central Pennsylvania, along with the grid layouts for the repeated GPR scans (with a blue bar illustrating survey lines used in analysis) and associated soil profiles with the locations of soil moisture sensors installed (white numbers indicate sensor depths and black numbers indicate soil horizon depths in meter).

contrast between two adjacent soil layers and an easier detection of the interface by GPR.

Topp et al. (1980) showed that soil dielectric constant ϵ can be expressed as a function of volumetric soil water content θ as:

$$\epsilon = 3.03 + 9.3\theta + 146.0\theta^2 - 76.7\theta^3. \quad (2)$$

For low-loss media (i.e., soils with low salinity and clay content), the propagation velocity of electromagnetic wave in the soil v can be related to the dielectric constant by

$$v = \frac{c}{\sqrt{\epsilon}}, \quad (3)$$

where c is free space electromagnetic velocity (0.3 m/ns).

Table 1
Soil profile characteristics of the two soil series investigated in this study.

Horizon	Depth (m)	Texture	Rock fragment ^a (%)	Bulk density (g cm ⁻³)	Pedality ^b (structure)	Roots
<i>Weikert series</i>						
Oe	0–0.05			0.425		Very many fine roots throughout
A	0.05–0.12	Silt loam	0	0.960	2 f gr	Very many fine roots throughout
Bw	0.12–0.28	Silt loam	60	1.260	2 f sbk	Common fine roots in cracks
CR	0.28–0.37	Silt loam	90		1 f sbk	Few fine roots in cracks
R	0.37					
<i>Rushtown series</i>						
Oe	0–0.05			0.425		Very many fine roots throughout
A	0.05–0.12	Silt loam	5		3 f gr	Very many fine roots throughout
Bw1	0.11–0.17	Silt loam	5	1.204	2 f sbk	Many fine roots throughout
Bw2	0.17–0.26	Silt loam	5	1.292	1 m sbk	Common fine roots throughout
Bw3	0.26–0.45	Silt clay loam	5	1.537	1 m sbk	Common fine to coarse roots throughout
BC	0.45–0.90	Silt clay loam	50	1.637	Massive	Few medium roots throughout
C	0.90–1.78		80	1.652	Massive	

^a Rock fragments are all shale channers of 2–150 mm thick, many of which are soft or very soft.

^b Pedality is described using ped grade, ped size, and ped shape: 1, 2, 3 for weak, moderate, and strong ped grade, respectively; vf, f, m, and c for very fine, fine, medium, and coarse ped size, respectively; gr and sbk for granular and subangular blocky ped shape, respectively. Massive means not soil structure developed.

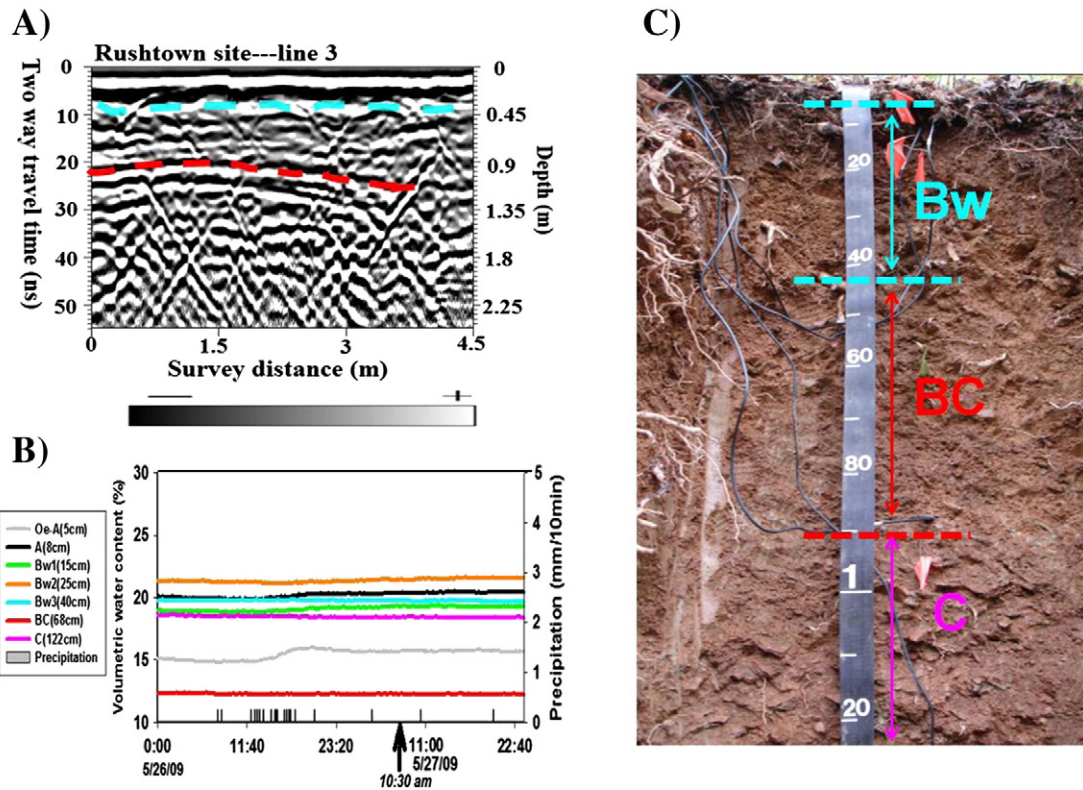


Fig. 2. A) Soil layer boundaries identified by GPR in the Rushtown soil (cyan dash line indicates interpreted Bw–BC interface and red dash line indicates interpreted BC–C interface; the depth was estimated using velocity = 0.09 m/ns). B) Real time soil water content monitoring when GPR survey was conducted in May 2009. C) Photograph of the Rushtown soil profile showing the Bw, BC, and C horizons.

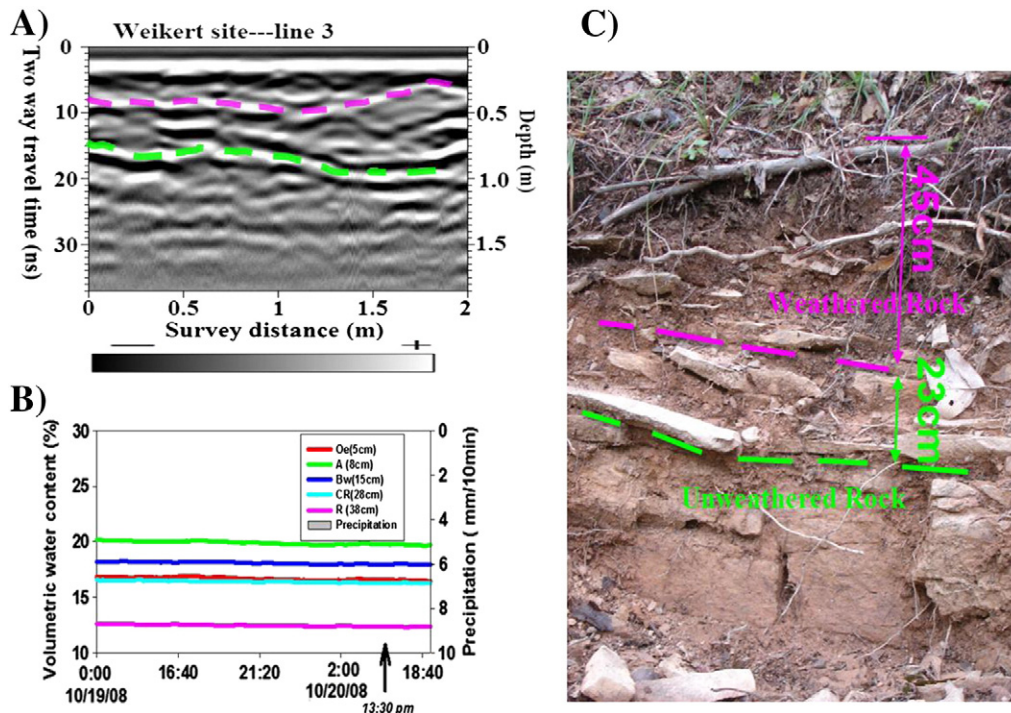


Fig. 3. A) Soil or rock layers identified by GPR in the Weikert soil (pink dash line indicates interpreted soil–bedrock interface and green dash line indicates interpreted weathered–unweathered rock interface; the depth was estimated using velocity = 0.1 m/ns). B) Real time soil moisture content monitoring when GPR survey was conducted in Oct. 2008. C) Photograph of the Weikert soil profile showing the soil–bedrock interface (pink dash line) and the weathered–unweathered rock interface (green dash line).

When the wave velocity is known, the depth of strong radar reflection d can be estimated by:

$$d = \frac{vt}{2}, \quad (4)$$

where t is two-way travel time of the radar signal from the antenna to the reflection point and back to the antenna.

Contrasting soil layers identified by GPR reflection were validated by direct field observations of soil profiles. First, we selected the largest peaks in the radargrams to identify the boundaries of layers. Then soil water contents measured by EC-5 sensors were used to estimate dielectric constant using Eq. (2). In the Rushtown soil, soil water content above the BC horizon was depth-averaged over the Oe, A, Bw1, Bw2, and Bw3 horizons. In the Weikert soil, soil water content above the R horizon was depth-averaged over the Oe, A, Bw, and CR horizons. After obtaining soil dielectric constant, the propagation velocity of electromagnetic wave was estimated using Eq. (3) and then the depth to the interface of contrasting layers was determined using Eq. (4). The identified

depths of reflections on the radargrams were compared with actual soil profiles to confirm our interpretations.

3. Results

3.1. Layer identification and validation

The identified layer boundaries for the Rushtown and Weikert soils are illustrated in Figs. 2 and 3, respectively. Here we selected only one line from each site to demonstrate layer boundaries because other survey lines showed similar patterns. In the Rushtown soil, GPR survey conducted in May 2009 showed a strong and continuous reflection at about 7–9 ns across survey line (cyan curve in Fig. 2A) and again at about 20–23 ns (red curve in Fig. 2A). By comparing soil water content data recorded at the same time when GPR surveys were conducted, we found that soil water content in the BC layer (red line in Fig. 2B) was distinctively lower than that in the soil horizons above and below it. We infer these reflections to correspond to the Bw–BC and BC–C

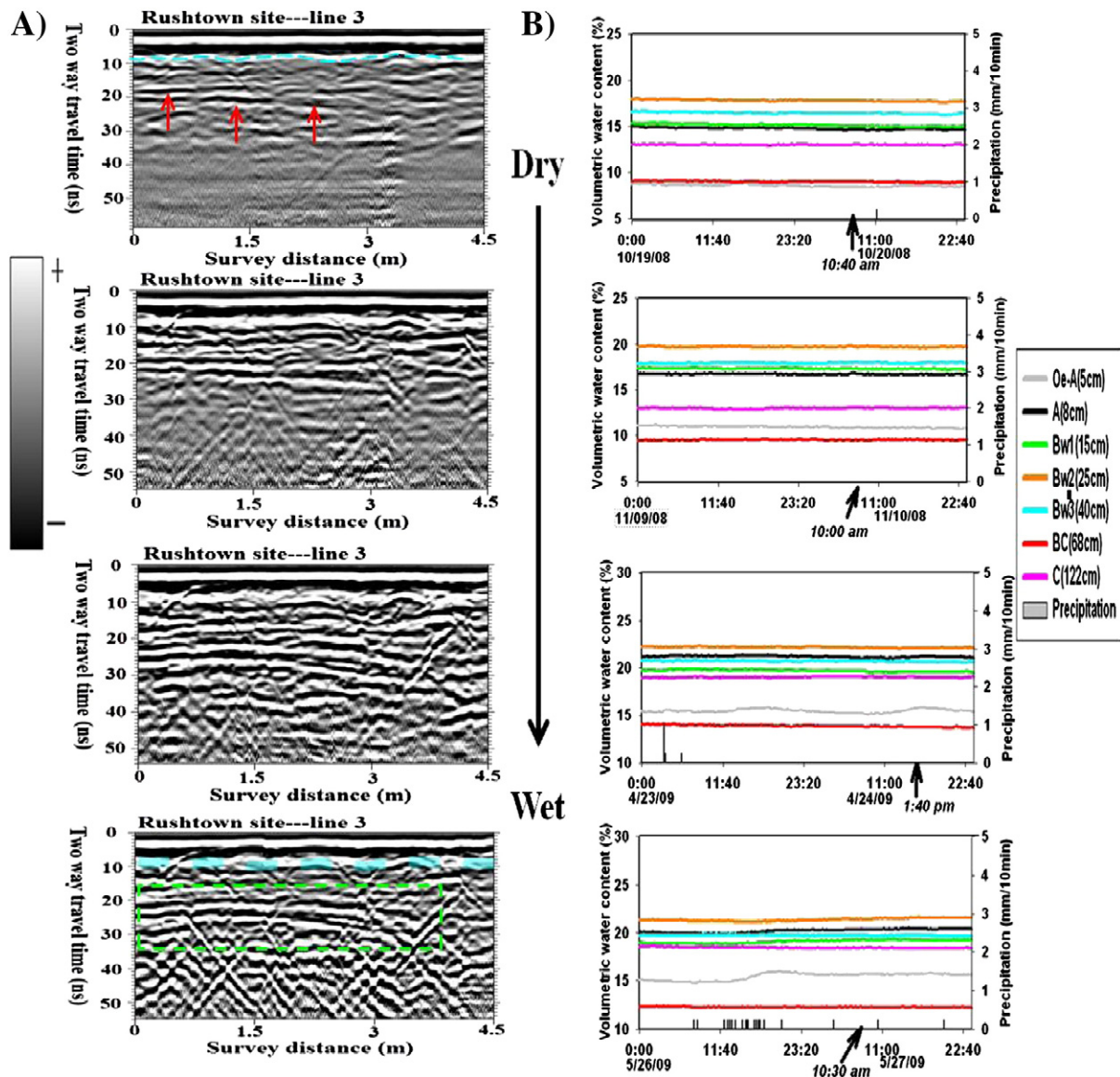


Fig. 4. A) Seasonal GPR signal strength changes in the deep Rushtown soil from dry to wet conditions (from top to bottom). Cyan dash lines indicate the Bw–BC interface, which enhanced the radar reflection as the soil became wet. Red arrows indicate intermittent reflection at the BC–C interface, which became more continuous as the soil became wet (as highlighted in green rectangular). B) Corresponding real-time soil moisture content monitoring at various depths during the GPR surveys (black arrows indicate the time when each GPR survey was conducted).

interfaces. To confirm this inference, dielectric constant and velocity above the Bw–BC interface was determined by applying Eqs. (2) and (3) using average soil water content above the BC layer. The average velocity was 0.09 m/ns. By applying Eq. (4), the estimated depths at the two strong reflections were about 32–41 cm and 90–104 cm, respectively, which were very close to the observed soil profile layering (Fig. 2C).

In the Weikert soil, two strong reflections were also observed in the radargrams at about 6–10 ns (pink line in Fig. 3A) and 15–18 ns (green line in Fig. 3A), respectively. By comparing with soil water content data, we believed the first reflection was the soil–bedrock interface (because soil water content at this interface was also distinctively lower than that in the soil horizons above). Using the methods mentioned above, the average velocity above the R horizon was 0.1 m/ns and the estimated depth of the first reflection were about 30–50 cm, which was consistent with the observed soil profile (pink line Fig. 3C). For the second strong reflection on the radargram, we interpreted it as the interface between weathered and unweathered rock, which is consisted of massive fractured shale bedrock as compared to the plate-like weathered rock in the R horizon (green line in Fig. 3C).

3.2. Seasonal GPR signal change in the deep Rushtown soil

Based on the identified layers in the radargrams, we investigated how these reflections changed seasonally in the two soils. Fig. 4A shows the GPR reflection profiles obtained in the same survey line for the four GPR surveys from October 2008 to May 2009, while Fig. 4B shows the corresponding soil moisture content changes when each of these GPR surveys was conducted. The soil water content showed a general dry to wet trend from October 2008 to May 2009. Fig. 5 summarizes soil water content change with time for the four GPR scans and the impact on reflection coefficient.

With increasing soil water content, the GPR reflection profiles show the following patterns:

- 1) The reflection at the Bw–BC interface (clear band at 10 ± 3 ns in all radargrams across the whole survey lines) became stronger as soil water content increased. This is illustrated using dash cyan line on the radargrams at the top and bottom in Fig. 4A. This stronger reflection can be explained by the increased reflection coefficient at the Bw–BC interface as indicated by cyan dots in Fig. 5. In addition, more hyperbolas and crisscross occurred near this interface. This

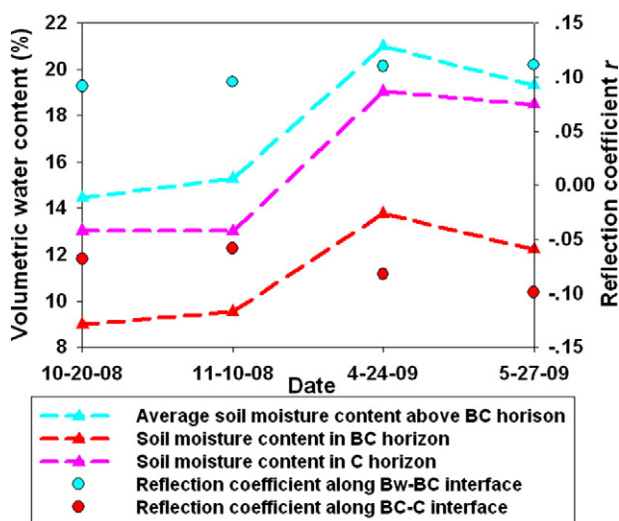


Fig. 5. Reflection coefficient changes with time in the Rushtown soil. Triangle dash lines indicate soil moisture contents in three horizons (Bw, BC, and C). Dots indicate reflection coefficients estimated using the corresponding soil water contents.

was obvious in the radargram conducted on May 27th, 2009 (i.e., the bottom radargram in Fig. 4A), which was conducted after a small rainfall event (Fig. 4B). The crisscross pattern and multiple hyperbolas dominated in the radargram.

- 2) The reflection at the BC–C interface (at 21 ± 3 ns in all radargrams across the survey lines from 0 to 3 m) became clearer and more continuous as soil became wet. On October 20th, 2008, several strong reflections intermittently showed at the BC–C interface (i.e., at 18–21 ns indicated by red arrows in top radargram in Fig. 4A), while on May 27th, 2009, the reflection at the BC–C interface (20–23 ns) displayed a stronger and linear pattern (i.e., the bottom radargram in Fig. 4A), and a continuous strong reflection was obvious between 0 and 3 m of the survey line. The clearer reflections at the BC–C interface can also be explained by reflection coefficient between the BC and C horizons (red dots in Fig. 5). Note that the reflection coefficients at the BC–C interface were negative values, which indicated that the dielectric constant in upper horizon was lower than that in the lower horizon. However, the absolute value of the reflection coefficient still increased as the soil became wet, indicating increased contrast between the BC and C horizons.

3.3. Seasonal GPR signal change in the shallow Weikert soil

Fig. 6 shows the seasonal GPR radargrams in the Weikert soil, along with the corresponding soil moisture content dynamics when the GPR surveys were conducted. With the increase of soil water content, the GPR reflection profiles show the following patterns:

- 1) Many hyperbolas intersected with the reflection at the soil–bedrock interface (at ~6–11 ns in all radargrams) as the soil became wet, which made the interface less continuous. This is obvious when comparing the radargram of October 2008 with that of April 2009. On October 20th, 2008, the radargram (top radargram in Fig. 6A) showed a clear and continuous reflection. In the first meter of the survey line, the reflection appeared at 8–10 ns; while in the remaining survey lines, the reflection gradually inclined to soil surface and reached to 6 ns at the end of the survey. However, the radargram on May 27th, 2009 showed intermittent strong reflection along the first meter of the survey line and several limbs of the hyperbolas broke the continuous reflection (red arrow in the bottom radargram in Fig. 6A).
- 2) The reflection at the weathered and unweathered bedrock interface (at ~15–23 ns in all radargrams) became less discontinuous along the interface and strong crisscross pattern occurred below the interface as the soil became wet, especially along the survey line of 0.5–1.5 m. When comparing radargrams under the dry condition (October and November 2008) with the wet condition (April and May, 2009), the radargrams under the dry condition (top radargram in Fig. 6A) showed a clear and continuous reflection at the interface while under the wet condition the linear reflection seemed broke along the 0.5–1 m survey line and a strong crisscross reflection occurred below (green rectangular in the bottom radargram in Fig. 6A).

4. Discussion

The amount of energy reflected back to a radar antenna is dependent upon the contrast of dielectric constant between two soil layers. The greater and/or more abrupt the contrast in the dielectric properties of adjoining layers, the greater the amount of energy would reflect back to the antenna, and the more intense and conspicuous the amplitude of the reflected signal would appear on radargrams (Doolittle et al., 2006). As the dielectric constant of soil materials is strongly dependent upon soil moisture content (Topp et al., 1980), seasonal changes in radar reflection on the identified soil layer interfaces can provide useful

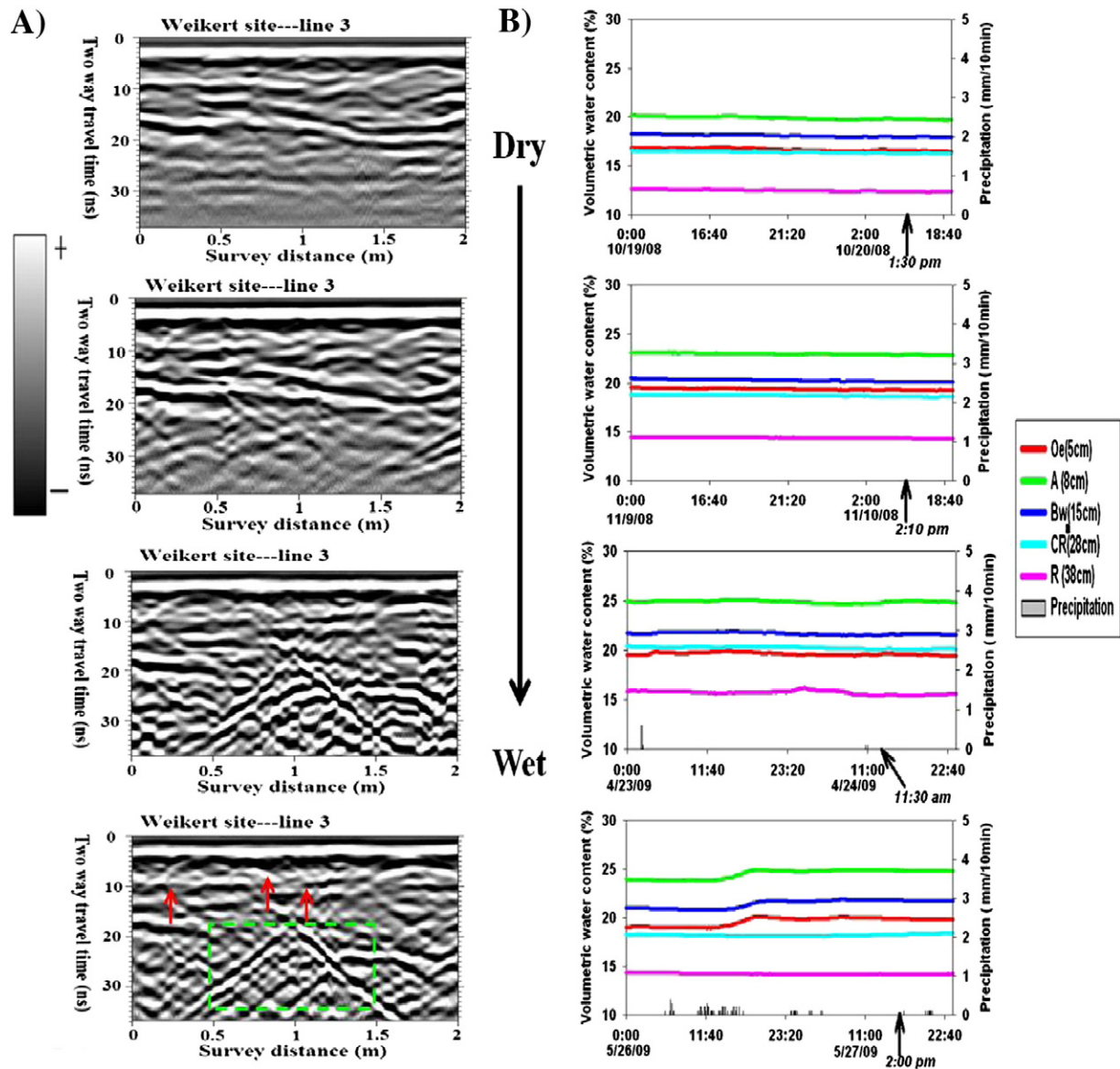


Fig. 6. A) Seasonal GPR signal strength changes in the shallow Weikert soil from dry to wet conditions (from top to bottom). Under dry condition (top radargram; same as Fig. 3A), two clear bands indicated the soil-bedrock interface and the weathered-unweathered rock interface (but we did not overlay interpreted lines at the interfaces to better show changes as soil became wet). Under wet condition (bottom radargram), these two bands were not continuous any more, and hyperbola and crisscrosses appeared. Red arrows indicated the presence of hyperbolas that broke the continuous reflection at the soil-bedrock interface. Green rectangular highlights the crisscrosses below the weathered-unweathered bedrock interface. B) Corresponding real-time soil moisture content monitoring at various depths during the GPR surveys (black arrows indicate the time when each GPR survey was conducted).

information about subsurface hydrologic processes and associated soil water distribution.

4.1. Implication for understanding hydrologic processes in the Rushtown soil

In the Rushtown soil, more hyperbolas and crisscross appeared in the radargrams as the soil became wet. These patterns have something to do with existing tree roots. Under the dry condition, the tree roots are too small to be detected by GPR (top radargram in Fig. 4A). However, under the wet condition, these tree roots formed preferential path for water redistribution and thus created dielectric contrast with the surrounding soil, leading to local diffractions in the radargrams. This effect has also been reported by Truss et al. (2007), who found that irregular rock surface and tree roots caused small scale preferential flow that led to multiple hyperbolic scatterings in GPR images.

Soil moisture content in the BC horizon (red line in Fig. 4B) was lower than that in the horizons above and below during the four GPR surveys from the dry to wet periods. This was also the case with long-term soil moisture monitoring data at this site (red line in Fig. 7A), indicating that the BC horizon was a water-restricting layer, which prevented water from further vertical infiltration. However, soil moisture monitoring data also showed that soil moisture content in the deeper C horizon (pink line in Fig. 7A) was higher than that in the BC horizon. This raised the question: how did water get to deeper C horizon to increase its moisture content? This scenario suggests that subsurface lateral flow had occurred from upslope areas that contributed to the increase in soil water content in the C horizon. Fig. 7B demonstrated a clear evidence of such lateral flow. Consequently, the linear reflection between the BC and C horizons under the wet condition was due to subsurface lateral flow into the C horizon, which enhanced the dielectric contrast across the BC-C interface so that the interface was clearly detected by GPR.

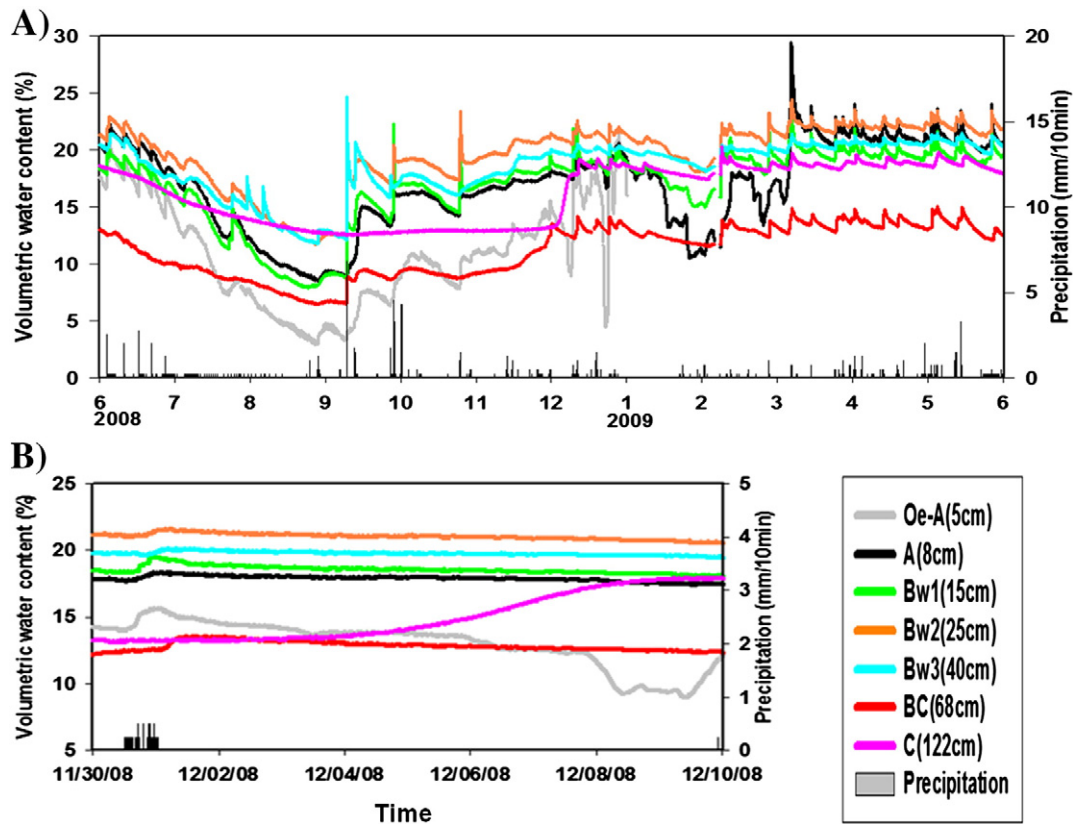


Fig. 7. A) Seasonal soil water content changes in the Rushtown soil from 2008 to 2009, indicating that soil moisture content in the BC horizon (red curve) was lower than that in the horizons above and below the BC. B) A demonstration of subsurface lateral flow from upslope area to increase soil moisture content in the C horizon (pink curve). After a rainfall event at the end of Nov. 2008, soil water content in the C horizon increased notably in the following several days, while the soil horizons above the C all showed an initial small spike during the rain, followed by a slow and steady decline after the rain. This indicates that vertical infiltration alone cannot support the increased amount of water in the C horizon, and subsurface lateral flow from upslope area must have occurred to supply extra water to the C horizon after the rain.

4.2. Implication for understanding hydrologic processes in the Weikert soil

In the Weikert soil, clear and linear reflections appeared at the soil–bedrock interface under the dry condition, while under the wet condition several limbs of hyperbolas intersected with the reflection and broke continuous reflection (Fig. 6A). The hyperbolas were caused by the water moving along tree roots in the upper soil horizon, which created hyperbolic scatterings in GPR images (as observed in the Rushtown soil).

In addition, water infiltrated into fractured bedrock can also lead to the diffraction of GPR wave, creating multiple hyperbolas and crisscross patterns at the interface between weathered and unweathered bedrock (bottom radargram in Fig. 6A). This implies another flow path in the shallow Weikert soil: under the wet condition, water can infiltrate into the fractures of underlying bedrock, and through slope gradient this water may move laterally to deeper soil layers in the downslope area.

4.3. Implication of seasonal GPR patterns in different soils

The seasonal changes in GPR signals in the two contrasting soils investigated in this study have demonstrated the effect of non-uniform soil water distribution on soil layer detection using GPR and a possible optimal time to apply GPR for identifying soil layers.

In the Rushtown soil, reflections at the Bw–BC and BC–C interfaces were more apparent under the wet condition than that under the dry condition. This was due to the presence of water-restricting BC horizon under the wet condition that accumulated water above the BC as well as soil moisture increase in the C horizon due to

subsurface lateral flow from the upslope area. This led to the dielectric contrast and thus GPR reflections in the Bw–BC and BC–C interfaces. However, GPR surveys conducted after rainfall also need to be considered because preferential flow along tree roots can cause multiple hyperbolas and crisscross patterns in the radargram. Isosurface (light red color in Fig. 8B), which extracted highest amplitude from volume data, was applied to highlight the BC–C interface in the Rushtown soil (Fig. 8B).

In the Weikert soil, reflections at the soil–bedrock interface and the weathered–unweathered bedrock interface were more obvious under the dry condition than that under the wet condition. This was due to infiltrating water moving along tree roots and into bedrock fractures, which created hyperbolas and crisscrosses along the interfaces and broke the continuous reflection. Isosurface (light red color in Fig. 8A) was applied to highlight the weathered–unweathered interface in the Weikert soil (Fig. 8A).

5. Summary and conclusions

Repeated GPR surveys were conducted during the dry to wet periods in two contrasting soils in the forested Shale Hills catchment. In the deep Rushtown soil located in the concave hillslope, the soil horizon interfaces of the Bw–BC and BC–C were identified by GPR. As the soil became wet, the Bw–BC interface showed more crisscross patterns and hyperbola shapes, indicating preferential water distribution. The BC–C interface showed more linear reflection under the wet than the dry conditions because of increased reflection coefficient due to water accumulation at that interface.

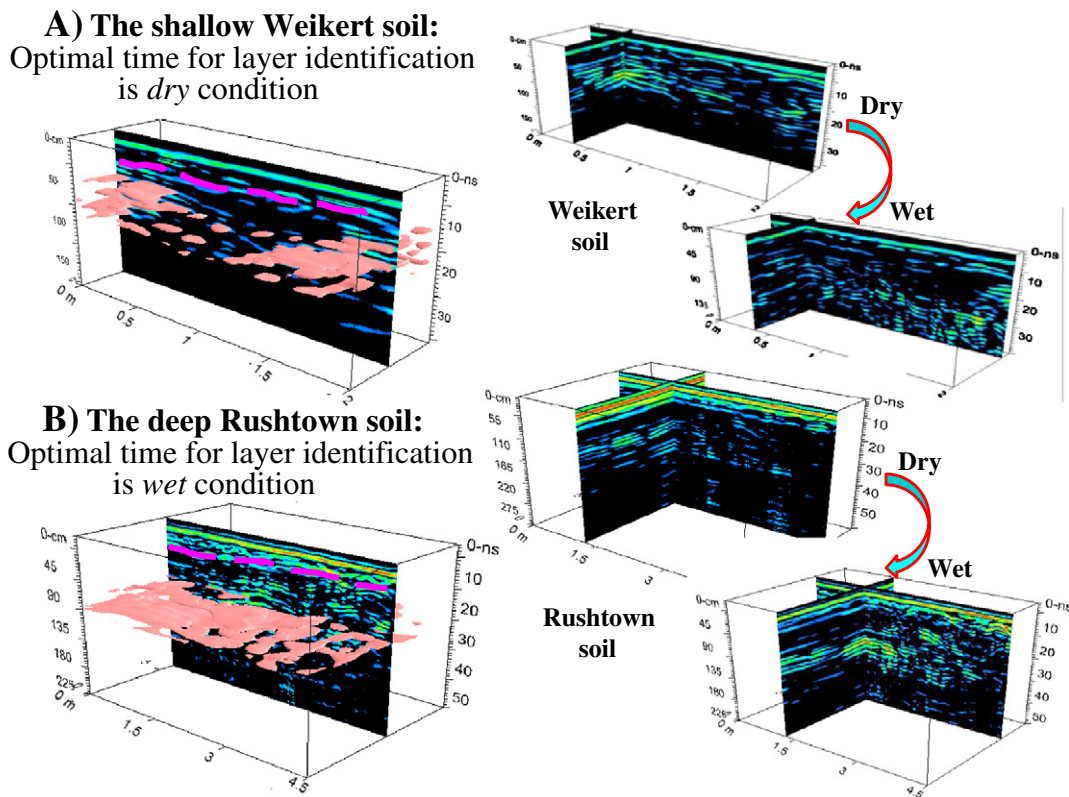


Fig. 8. 3D rendering of GPR images from dry to wet conditions in the Weikert and Rushtown soils, showing their different optimal times for identifying soil and/or rock layer interfaces using GPR: A) layer identification in the Weikert soil is optimal under dry condition. Pink dash line indicates the soil–bedrock interface, while light red color is isosurface of amplitude volume to highlight the weathered–unweathered bedrock interface; B) Layer identification in the Rushtown soil is optimal under wet condition. Pink dash line indicates the Bw–BC interface, while light red color is isosurface of amplitude volume to highlight the BC–C interface.

In the shallow Weikert soil located in the planar hillslope, the soil–bedrock interface and weathered–unweathered rock interface showed a clear linear reflection in the GPR radargram during the dry period. However, during the wet period, the soil–bedrock interface showed intermittent local strong reflection that was caused by preferential water distribution in the fractured shale bedrock. Near the weathered–unweathered interface, strong crisscross patterns occurred under the wet condition, reflecting non-uniform water distribution inside the rock.

The seasonal GPR signal changes suggest that GPR can be used to detect subsurface flow dynamics between dry and wet periods and that there is an optimal time for using GPR to detect soil and/or rock layers. Water-restricting BC layer and subsurface lateral flow were identified in the Rushtown soil during the wet period by recognizing the increase in radar reflection at the Bw–BC and BC–C interfaces. Water infiltrating into the fractured bedrock in a non-uniform pattern was clearly captured in the shallow Weikert soil under the wet condition because of abrupt dielectric contrast between water and the fractured rock. In the deep Rushtown soil, wet period without rainfall event was an optimal time for identifying water-restricting soil layers using GPR; whereas in the shallow Weikert soil, dry condition was more suited for identifying the soil–bedrock and weathered–unweathered rock interfaces using GPR.

Overall, this study has demonstrated the value of repeated GPR surveys to identify subsurface hydrologic processes and the advantages of combining real-time soil moisture monitoring with repeated GPR surveys to reveal non-uniform moisture distribution in soils and fractured rocks.

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